

Technical Review of the KBMI 4 Green-Financing Manuscript

Paper-review audit

30 August 2026

Overall assessment

The manuscript addresses a relevant question and presents an unusually complete-looking set of tables and appendices. However, its central empirical evidence is not reproducible from the data printed in Appendix 1, and several printed observations conflict with official bank disclosures. Consequently, the reported coefficients, significance tests, diagnostic conclusions, causal claims, and policy recommendations cannot presently be relied upon. This is a major data-integrity and methods problem, not a copyediting problem. The manuscript requires reconstruction of the dataset and complete regeneration of Chapter 4 before its substantive conclusions can be assessed.

The review inspected the full manuscript, equations, panel formulas, all 80 appendix rows, reported EViews output, hypotheses, conclusions, implementation-facing syntax, regulatory statements, citation integrity, and references. Sign conventions, symbol definitions, dimensions, branch conventions, and implementation ambiguity were checked explicitly. The manuscript contains no inverse-trigonometric or coordinate-transform formulas, so no branch or quadrant error arises in those categories.

Priority technical findings

1. **Definite error:** Appendix 1 does not reproduce any of the main empirical output

Location. Chapter 4, Tables 4.1, 4.4a, 4.4b, and 4.5; Appendix 1; Appendix 2.

Problem. Recalculation from the 80 unique rows printed in Appendix 1 produces materially different descriptive statistics and correlations. Selected comparisons are:

Quantity	Manuscript	Recalculated from Appendix 1
Mean Green Financing	23.85	23.45875
Mean CAR	22.64	23.46125
Mean ROA	3.18	3.28225
Median ROA	3.12	3.29
$\text{corr}(ROA, GF)$	0.5824	0.8628
$\text{corr}(ROA, NPL)$	-0.6412	-0.8698
$\text{corr}(ROA, CAR)$	0.5120	0.8409
$\text{corr}(NPL, CAR)$	-0.3650	-0.9340

An ordinary least-squares dummy-variable reproduction of the stated bank fixed-effects model gives

$$\widehat{ROA}_{it} = \hat{\alpha}_i + 0.07671 GF_{it} - 0.07212 NPL_{it} + 0.08129 CAR_{it},$$

with conventional non-robust p -values below 0.0001, 0.131, and below 0.0001, respectively; $R^2 = 0.9866$, adjusted $R^2 = 0.9855$, and Durbin–Watson = 0.566. The manuscript instead reports slopes 0.04218, -0.38542 , and 0.07415 , adjusted $R^2 = 0.7026$, and Durbin–Watson = 1.942. Applying the manuscript’s own printed coefficients and bank effects to Appendix 1 gives residual sum of squares 6.9878, not the reported 6.3601.

Why it matters. The discrepancy overturns at least Hypothesis 2: in the direct reproduction, the NPL slope is not significant at the 5% level. It also means that the EViews blocks in Appendix 2 cannot have been generated from Appendix 1 as printed. Every results-dependent statement in the abstract, Chapter 4, Chapter 5, and the policy roadmap is therefore unsupported.

Suggested fix. Rebuild one machine-readable dataset directly from traceable source documents; preserve every source page, reporting basis, unit, and transformation; import that exact file into EViews; and regenerate every table and appendix automatically. Include the EViews workfile or executable script and an observation-level provenance sheet. Do not revise isolated numbers by hand.

2. **Definite error:** Printed observations conflict with official financial disclosures

Location. Chapter 4 bank tables and Appendix 1, especially BBRI 2025-Q1 through 2025-Q4.

Problem. The BBRI 2025-Q4 row reports CAR 26.20%, gross NPL 2.10%, and ROA 4.15%. BRI’s official financial publication for 31 December 2025 reports CAR 21.06%, gross NPL 3.29%, and ROA 3.26% under its displayed bank performance-ratio basis; it also reports a separate ROA measure of 2.64%. The official first-quarter 2025 publication reports gross NPL 3.14% and ROA 2.95%, whereas Appendix 1 gives 2.25% and 3.88%.¹

Why it matters. This is not explainable by rounding. If the manuscript uses consolidated rather than bank-only figures, a different ROA convention, or a derived measure, that basis is not identified and still does not reconcile the displayed values. The discrepancy raises a serious provenance concern for all 80 rows.

Suggested fix. Replace every observation from the original filings and state, for each variable, whether it is bank-only or consolidated, period-end or period-average, quarterly or year-to-date, and annualized or unannualized. Add a source-page column and retain a reconciliation record for any derived ratio.

3. **Definite error:** Quarterly Green Financing data have no documented construction rule

Location. Sections “Sources of Data” and “Operationalization of Variables”; Tables 4.1b–4.1e; Appendix 1.

Problem. The manuscript says that Green Financing is obtained from annual sustainability reports, yet supplies a different smoothly changing value for every quarter. It does not state whether

¹PT Bank Rakyat Indonesia (Persero) Tbk, official quarterly financial publications for 31 March 2025 and 31 December 2025, performance-ratio tables, accessed 30 August 2026 through BRI Investor Relations.

annual values were repeated, interpolated, reconstructed from quarterly balances, or combined with another source. It also alternates among “green financing,” KUBL, and broader sustainable financing without specifying which categories enter the numerator.

Why it matters. Interpolation or unreported allocation creates artificial within-bank variation and can mechanically generate the strong trends used by the regression. Broader sustainable-finance totals may include social or MSME financing and are not interchangeable with a green-credit numerator.

Suggested fix. Define the numerator and denominator for every year under the applicable taxonomy, give the exact source table for every observation, and use annual panel data if only annual Green Financing is observed. If a quarterly series is genuinely available, document the source and do not interpolate silently.

4. **Unsupported claim:** The design does not identify causal effects

Location. Research design; hypothesis tests; discussion; abstract and conclusions.

Problem. The manuscript describes a causal design and repeatedly says that the regressions “prove” that Green Financing, NPL, and CAR determine ROA. The model uses contemporaneous ratios, four selected banks, bank fixed effects only, and no identification strategy. Green Financing, risk, capital, and profitability can be jointly determined; profitable banks may expand green lending or capital, and common macroeconomic time trends affect all variables.

Why it matters. The appendix series are dominated by common monotonic trends: Green Financing, CAR, and ROA generally rise while NPL falls. Bank fixed effects remove time-invariant bank differences but do not remove common time shocks or trends. A coefficient therefore cannot be interpreted as a causal effect or as evidence for the proposed funding-cost and debtor-quality mechanisms.

Suggested fix. Either reframe the study consistently as associational or introduce a defensible identification design. At minimum, estimate bank and quarter fixed effects, define lags, add justified controls, investigate reverse causality, and report sensitivity to time trends. Mechanism claims require measured mediators or should be presented as hypotheses rather than findings.

5. **Definite error:** The diagnostic conclusions are incompatible with the printed data and use unsuitable tests

Location. Sections on stationarity and classical assumptions; Tables 4.2 and 4.4; Appendix 2.

Problem. Direct reconstruction gives a Durbin–Watson statistic near 0.566, with within-bank lag-one residual correlations approximately 0.61–0.91, not the reported 1.942. The manuscript applies a conventional Durbin–Watson interval to stacked panel residuals and does not test cross-sectional dependence. It also reports LLC/IPS stationarity at level despite visibly trend-dominated series, without providing settings, lags, deterministic terms, or raw output. Finally, normal residuals are described as a requirement for BLUE, which they are not.

Why it matters. Serial correlation, cross-sectional dependence, and common shocks can make the reported standard errors and p -values severely misleading. With only four bank clusters, ordinary cluster-robust asymptotics also require caution.

Suggested fix. After correcting the data, report a panel serial-correlation test, a cross-sectional-dependence diagnostic, and inference robust to the supported error structure. State unit-root specifications completely and test sensitivity to deterministic trends. Avoid claiming that all assumptions are “perfectly” satisfied.

6. **Definite error:** The formulas and interpretations for adjusted R^2 and the FEM F test use the wrong degrees of freedom

Location. Section “Model Feasibility and Hypothesis Testing” and the interpretation of Hypothesis 4 and adjusted R^2 .

Problem. With $NT = 80$, $N = 4$ bank intercepts, and $K = 3$ slopes, the printed formula

$$1 - \frac{(1 - R^2)(NT - 1)}{NT - K - 1}$$

ignores the bank effects. Substituting the reported $R^2 = 0.72514$ gives 0.71429, not the reported 0.70258. The value 0.70258 is obtained when the denominator reflects all estimated parameters. Likewise, the standard overall FEM F statistic tests the slopes together with the non-reference bank effects; it is not automatically a test that only GF , NPL , and CAR are jointly zero.

Why it matters. Hypothesis 4 is not established by the reported statistic, and the model R^2 cannot be described as the proportion explained solely by the three substantive predictors because it also contains bank effects.

Suggested fix. Use the correct residual degrees of freedom $NT - N - K$ for the stated parameterization. Report a separate Wald test of $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$, with its numerator and denominator degrees of freedom. Distinguish overall, within, between, and incremental explanatory power.

7. **Definite error:** Required notation is corrupted and key symbols change meaning

Location. Variable formulas in Chapter 2; panel equations and matrix algebra in Chapter 3; Tables 4.2, 4.5, and 4.6; elasticity section; Appendix 4.

Problem. The character “?” replaces multiplication signs, inequalities, Greek letters, `\left`, `\multirow`, and `\multicolumn` throughout the manuscript. More importantly, N is first defined as the number of banks ($N = 4$) but is used in the Jarque–Bera formula as if it were the number of observations. K is first the number of regressors ($K = 3$) but becomes kurtosis in the same formula. In the fixed-effects discussion, the common intercept, bank-specific intercept, and bank deviation are visually collapsed into indistinguishable corrupted symbols.

Why it matters. A literal implementation can substitute $N = 4$ and $K = 3$ into the Jarque–Bera formula and obtain the wrong statistic. The damaged panel notation prevents an unambiguous distinction among α , α_i , μ_i , and β .

Suggested fix. Restore the source encoding globally, then use $n = NT$ for sample size, N for banks, T for periods, K for slopes, and κ for kurtosis. Define, for example, $\alpha_i = \alpha + \mu_i$ and keep those verbal descriptors unchanged everywhere.

8. **Definite error:** The manuscript’s multicollinearity claim is contradicted by Appendix 1

Location. Table 4.4a, Table 4.4, and the accompanying discussion.

Problem. The manuscript reports the largest absolute correlation among regressors as 0.4120 and centered VIFs of 1.28–1.42. Appendix 1 instead gives $\text{corr}(NPL, CAR) = -0.9340$. A pooled calculation gives VIF approximately 10.05 for NPL and 8.31 for CAR; within-bank VIFs are approximately 7.0–7.3.

Why it matters. The manuscript’s assertion that multicollinearity is absent is not supported. The highly synchronized trends make individual slope estimates unstable and complicate claims about which variable is most important.

Suggested fix. Recompute correlations and VIFs from the final estimation sample and report them under the same transformation used by the estimator. Address common trends and report coefficient stability across plausible specifications.

9. **Definite error:** Raw coefficient magnitudes and elasticities are interpreted incorrectly

Location. Hypothesis 2 discussion and “Partial Sensitivity and Elasticity Analysis.”

Problem. The manuscript calls NPL the most sensitive determinant because $|-0.385|$ is the largest raw coefficient, although the regressors have different distributions. Later, its own reported elasticities rank CAR above Green Financing and NPL. The elasticity values are dimensionless, but are printed with percent signs and described as if a 1% relative increase and a one-percentage-point increase were interchangeable.

Why it matters. This yields contradictory substantive rankings and confuses marginal effects with elasticities.

Suggested fix. Do not rank importance from raw slopes. State that a one-percentage-point change has the slope’s percentage-point association with ROA. For elasticity, state that a 1% relative change in X_j is associated with an estimated $\eta_j\%$ relative change in ROA at the evaluation point, and do not append a percent sign to η_j itself.

10. **Likely issue:** PSAK 71/IFRS 9 treatment is over-simplified and partly mis-stated

Location. Section “Asset Quality, Collectibility, and Implementation of PSAK 71/IFRS 9” and the NPL discussion.

Problem. The text maps regulatory collectibility mechanically onto IFRS 9 stages and suggests that Stage 3 requires provisioning to 100% of unsecured exposure. The displayed ECL equation omits scenario probabilities and does not clarify whether PD_t is marginal, conditional, or cumulative. IFRS 9 recognizes probability-weighted discounted cash shortfalls and does not impose a universal 100% allowance merely because an asset is Stage 3.

Why it matters. The description can mislead readers about the mechanism connecting NPL and CKPN and is too coarse for an implementation-facing formula.

Suggested fix. Check the wording against PSAK 71/IFRS 9, distinguish prudential collectibility from accounting staging, define all ECL terms and probability conventions, and label any reduced formula explicitly as a simplification.

11. **Definite error:** Regulatory descriptions contain checkable inaccuracies

Location. Chapter 1 regulatory background; Chapter 2 KUBL list; Appendix 5.

Problem. POJK 51/POJK.03/2017 does require sustainability reporting, but its reporting outline lists independent written verification “if any”; it does not make independent audit universal in the way the manuscript states. The manuscript presents 11 KUBL categories and omits the MSME category included in the relevant sustainable-business classification. It also treats POJK 60/POJK.04/2017 as the operative green-bond rule without noting its replacement by POJK 18 of 2023.²

Why it matters. These errors affect the definition of the Green Financing numerator and the manuscript’s policy narrative.

Suggested fix. Quote the operative provisions precisely, distinguish KUBL from the broader sustainable-business categories, state the classification used for each sample year, and update superseded regulations.

12. **Definite error:** Citation and reference integrity is not adequate

Location. Literature review tables, in-text parenthetical keys, and “References.”

Problem. No `references.bib` or rendered reference list is present, although the manuscript ends with `\bibliography{references}`. The strings such as (Pramono2022) are not valid LaTeX citation commands. Spot checks also reveal title/content mismatches: Cui et al. (2018) is *The Impact of Green Lending on Credit Risk in China*, not the title printed in Table 2.1; Chiaramonte et al. (2020) is an insurance-industry article, not the bank study claimed there; and the Yin et al. (2021) article has a different title and materially more heterogeneous findings than the summary reports.³

Why it matters. The claimed research gap and theoretical support cannot be evaluated when citations are absent, misidentified, or summarized inaccurately.

Suggested fix. Build a complete bibliography from DOI/publisher metadata, convert every citation to `\cite` or the chosen author–year command, and verify each table row against the actual article’s sample, outcome, method, and conclusion.

²Otoritas Jasa Keuangan, POJK 51/POJK.03/2017 and its reporting appendix; OJK summary of POJK 18/2023; official TKBI publications, accessed 30 August 2026.

³Yin et al., “The determinants of green credit and its impact on the performance of Chinese banks,” *Journal of Cleaner Production* (2021); Cui et al., “The Impact of Green Lending on Credit Risk in China,” *Sustainability* 10(6):2008 (2018); Chiaramonte et al., “Sustainability Practices and Stability in the Insurance Industry,” *Sustainability* 12(14):5530 (2020).

Meaningful editorial and presentation findings

- **Repeated overstatement across the abstract, results, and conclusion.** Replace “proves,” “establishes the validity of a theory,” “resolves the academic debate,” and similar causal language with wording proportional to the corrected design, such as “is associated with” or “is consistent with.”
- **Unsupported policy precision in Chapter 5.** Targets such as CAR of 22–26%, Green Financing of 32–40%, and green-asset risk weights of 50–75% do not follow from the regression. Either provide a separate calibration and prudential analysis or remove the numeric prescriptions.
- **Fixed-effect intercepts are over-interpreted.** A fitted intercept at $GF = NPL = CAR = 0$, far outside the observed sample, does not demonstrate CASA strength, efficiency, or resilience. Describe bank effects as conditional intercept differences only.
- **Quarterly measurement requires consistent bases.** Clarify whether ROA is an annualized year-to-date regulatory ratio, whether NPL and CAR are period-end stocks, and whether all variables use bank-only or consolidated reporting. Mixing these bases invalidates coefficient interpretation.

Proofing sweep

The bundled proofing scanner returned nine SEMICOLON_CAP candidates. All were false positives after inspection because the capitalized tokens were acronyms or bank codes; none should be changed solely because of that rule. The following manual hits are high-confidence defects:

- **Title/front matter:** “2021?2025” and “Bab 1?5” should be restored to “2021–2025” and “Bab 1–5.” Confirm whether “CFP?” is intended to contain a registered-mark symbol.
- **Throughout equations and tables:** restore every corrupted “?” token, including $\times$, $\geq$, $\leq$, α , β , η , `\left`, `\multirow`, and `\multicolumn`.
- **Chapter 4 headings:** “Analisis Tren Kuartalan dan Profil Perbankan...” is immediately duplicated by “Analisis Tren Kuartalan dan Profil Finansial Perbankan...”; retain one heading.
- **Chapter 4 prose:** change “batas batas toleransi” to “batas toleransi” and “antarbang” to “antarbank.”
- **Headings and PDF strings:** many headings split an opening italic marker from its closing parenthesis, and the adjusted- R^2 `\texorpdfstring` construction has malformed braces. Repair these before conversion to LaTeX.
- **Table 5.2:** the Treasury row contains an unescaped column-separating ampersand between *Green Bond* and *Sustainability-Linked Sukuk*, shifting the remaining cells.
- **Front matter:** replace all placeholders for NIM, supervisor, examination date, signatures, and curriculum-vitae details before submission.
- **Statistical reporting:** replace printed $p = 0.0000$ with a bound such as $p < 0.0001$.

Highest-priority revision sequence

1. Suspend all results, conclusion, and policy text; reconstruct and document the raw dataset from official filings.
2. Regenerate the entire empirical analysis from that one dataset and attach reproducible EViews commands/output.
3. Re-specify the panel model to address time effects, serial dependence, common shocks, and endogeneity; otherwise limit claims to association.
4. Correct the degrees of freedom, hypothesis tests, coefficient/elasticity interpretations, and notation.
5. Rebuild and verify the bibliography and update regulatory statements.
6. Only after the technical results stabilize, complete the bounded language and typesetting corrections.

Items requiring external verification

Every one of the 80 observations should be checked against the original bank filing, not merely against a secondary compilation. The precise annual Green Financing numerator for each bank and year needs verification because sustainability reports use differing labels and scopes. The manuscript's cited Indonesian studies whose exact titles could not be confirmed from the supplied files also require DOI, journal, volume, issue, page, and result verification. Finally, the current OJK/BI rules and taxonomy versions should be checked as of the manuscript's August 2026 submission date; the empirical classification used for 2021–2025 must remain period-appropriate and must not be retroactively mixed without explanation.